

NAME:

Total Marks: 34

Time Allowed: 45 minutes

1. (6 marks)

- (i) Use the following data to calculate the binding energy per nucleon in MeV/nucleon for Beryllium-9: [4]

$$\text{Mass of } {}^9_4\text{Be} = 9.012\,183\,\text{u} \quad (\text{excludes electrons})$$

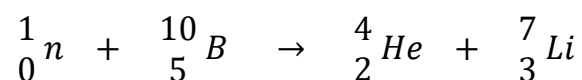
$$\text{Mass of proton} = 1.007\,276\,\text{u}$$

$$\text{Mass of neutron} = 1.008\,665\,\text{u}$$

- (ii) The binding energy per nucleon for Beryllium-10 is 6.29 MeV/nucleon. Which of the two isotopes of Beryllium is more stable, and why? [2]

2. (10 marks)

Boron neutron-capture therapy is used in the treatment of cancerous tumours. When Boron-10 captures a low-energy neutron, it undergoes the reaction



- (i) Use the following data to determine the energy released in joules for the above reaction. [4]

$$\text{Mass of } {}_5^{10}B = 10.012\,937\text{ u}$$

$$\text{Mass of } {}_2^4He = 4.002\,603\text{ u}$$

$$\text{Mass of } {}_3^7Li = 7.016\,004\text{ u}$$

$$\text{Mass of neutron} = 1.008\,665\text{ u}$$

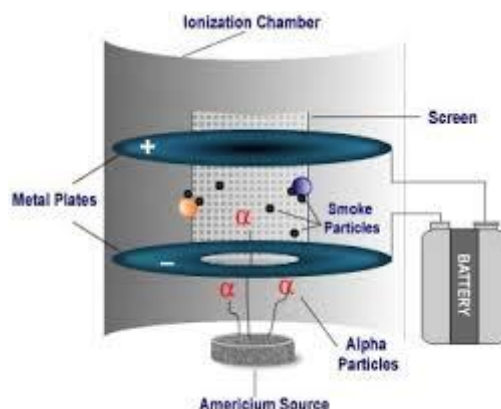
The neutrons used are slow neutrons, each of energy $8.00 \times 10^{-20} \text{ J}$.

- (ii) If 1000 slow neutrons miss the tumour and are absorbed by 5.00 g of healthy tissue, what dose equivalent does the patient receive? [3]

- (iii) How many slow neutrons absorbed by 1.00 kg of healthy tissue would produce a massive dose of 1.00 Sv? [3]

3. (14 marks)

Some smoke detectors use the electrical properties of ionised air molecules and the changes brought about when smoke particles enter the device. These detectors typically use Americium-241 to emit α -particles.



Suppose a new device contains $0.300\ \mu\text{g}$ (note $1\ \text{microgram}\ (\mu\text{g}) = 1 \times 10^{-6}\ \text{g}$) of Americium-241 with an initial activity of $37.0\ \text{kBq}$. Americium-241 has a half-life of 432 years and an atomic mass of $241.056\ 829\ \text{u}$ (including the electrons).

- (i) Write down the equation for the α -emission from Americium-241, clearly identifying the daughter nucleus. [2]
- (ii) Why is the α -radiation unlikely to escape from the device? [1]
- (iii) What will the activity in such a device be after 43.2 years? [3]

- (iv) How many molecules of Americium-241 does the device initially contain? [4]
- (v) How many molecules of Americium-241 does the device contain after 43.2 years? (should you not get an answer for iv, assume the result is 7.5×10^{14}) [2]
- (vi) How many α -particles are emitted in 43.2 years? [2]

4. (4 marks)

Geiger counters and cloud chambers both rely on the fact that nuclear radiation creates ions.

(i) Briefly explain how ions are created by nuclear radiation. [2]

(ii) Place in order from highest to lowest the ionising ability of α , β and γ particles: [1]

1.
2.
3.

(iii) Place in order from highest to lowest the maximum speed of α , β and γ particles: [1]

1.
2.
3.

- End of Questions -